Class 8: Mini Project

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Preparing the data

We need to read in the files.

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
```

Examine the data to make sure the column names are correct.

```
head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
```

```
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

The first column tells us which cells are malignant. We don't want this in the analysis, so we'll delete the first column.

```
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
```

We'll set up a new vector called diagnosis that contains the data from the diagnosis column. We'll store it as a factor and use it later to check our results.

```
# Create diagnosis vector for later
diagnosis <- wisc.df$diagnosis
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```r
nrow(wisc.df)
```

```
[1] 569
```

This dataset has 569 observations.

> Q2. How many of the observations have a malignant diagnosis?

```r
table(diagnosis)
```

```
diagnosis
  B   M 
357 212 
```

212 observations have a malignant diagnosis.

> Q3. How many variables/features in the data are suffixed with `_mean`?

```r
length(grep("_mean", colnames(wisc.df)))
```

```
[1] 10
```

10 variables in the data are suffixed with `_mean`.

Principal Component Analysis

First check if the data needs to be scaled before performing PCA.

We will check the mean and standard deviation of the features of the data to determine this.

```r
# Check column means and standard deviations
colMeans(wisc.data)
```

```
           radius_mean            texture_mean          perimeter_mean
          1.412729e+01            1.928965e+01            9.196903e+01
             area_mean         smoothness_mean        compactness_mean
          6.548891e+02            9.636028e-02            1.043410e-01
        concavity_mean     concave.points_mean           symmetry_mean
          8.879932e-02            4.891915e-02            1.811619e-01
 fractal_dimension_mean               radius_se              texture_se
          6.279761e-02            4.051721e-01            1.216853e+00
          perimeter_se                 area_se           smoothness_se
          2.866059e+00            4.033708e+01            7.040979e-03
        compactness_se            concavity_se       concave.points_se
          2.547814e-02            3.189372e-02            1.179614e-02
           symmetry_se    fractal_dimension_se            radius_worst
          2.054230e-02            3.794904e-03            1.626919e+01
         texture_worst         perimeter_worst              area_worst
          2.567722e+01            1.072612e+02            8.805831e+02
      smoothness_worst       compactness_worst         concavity_worst
          1.323686e-01            2.542650e-01            2.721885e-01
  concave.points_worst          symmetry_worst fractal_dimension_worst
          1.146062e-01            2.900756e-01            8.394582e-02
```

```r
apply(wisc.data,2,sd)
```

```
           radius_mean            texture_mean          perimeter_mean
          3.524049e+00            4.301036e+00            2.429898e+01
             area_mean         smoothness_mean        compactness_mean
          3.519141e+02            1.406413e-02            5.281276e-02
        concavity_mean     concave.points_mean           symmetry_mean
          7.971981e-02            3.880284e-02            2.741428e-02
 fractal_dimension_mean               radius_se              texture_se
          7.060363e-03            2.773127e-01            5.516484e-01
          perimeter_se                 area_se           smoothness_se
          2.021855e+00            4.549101e+01            3.002518e-03
        compactness_se            concavity_se       concave.points_se
          1.790818e-02            3.018606e-02            6.170285e-03
           symmetry_se    fractal_dimension_se            radius_worst
          8.266372e-03            2.646071e-03            4.833242e+00
         texture_worst         perimeter_worst              area_worst
          6.146258e+00            3.360254e+01            5.693570e+02
      smoothness_worst       compactness_worst         concavity_worst
          2.283243e-02            1.573365e-01            2.086243e-01
  concave.points_worst          symmetry_worst fractal_dimension_worst
```

```
6.573234e-02            6.186747e-02            1.806127e-02
```

Run PCA with `prcomp()` on `wisc.data`, scaling if appropriate, and assign the output to `wisc.pr`.

```r
# Perform PCA on wisc.data by completing the following code
wisc.pr <- prcomp(wisc.data, scale=TRUE)
```

Look at a summary of the results using `summary()`.

```r
# Look at summary of results
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

44.27% of the original variance is captured by PC1.

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 principal components are required to describe at least 70% of the original variance in the data.

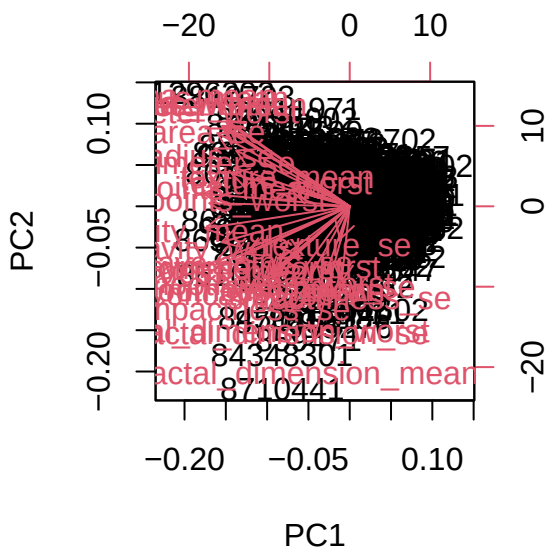
> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 principal components are required to describe at least 90% of the original variance in the data.

Interpreting PCA results

We'll first create a biplot of `wisc.pr` using `biplot()`.

```
biplot(wisc.pr)
```



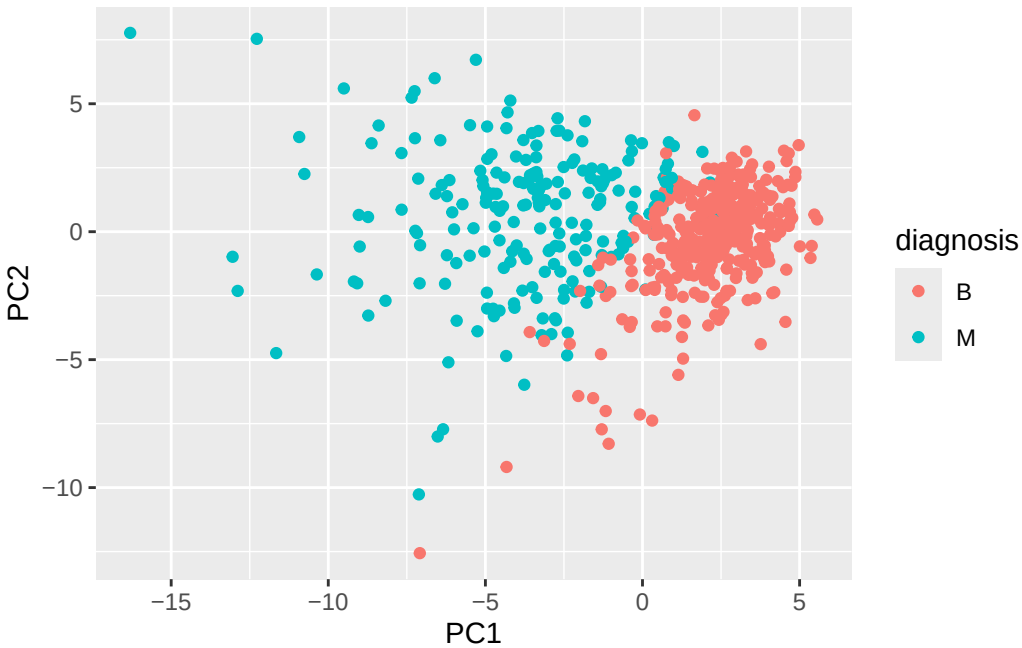
> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot is almost incomprehensible with the number of points and labels on it.

Instead, let's create a scatterplot of PC1 vs PC2 and color the points by diagnosis.

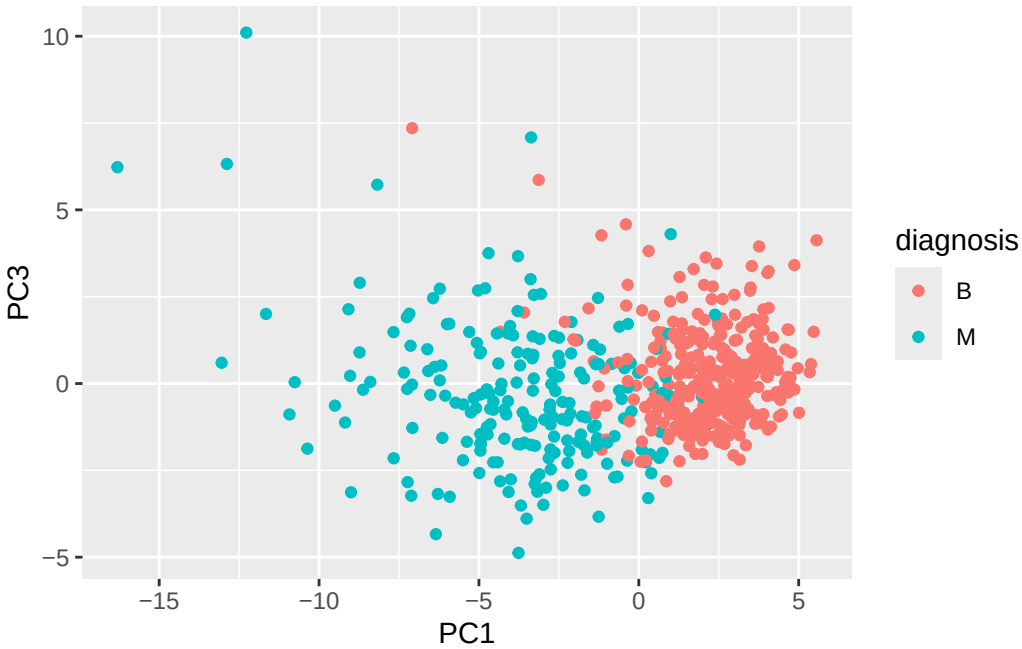
```r
# Scatter plot observations by components 1 and 2
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```r
# Repeat for components 1 and 3
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



Both plots show a separation between benign and malignant based on PC1.

Variance explained

A scree plot can be used to show the variance that each PC captures. If the plot shows an elbow, that will determine how many PCs to use for further analysis. If there is no elbow, a threshold like 70% or 90% variance is used instead.

We will calculate the variance of each PC by squaring the `sdev` component of `wisc.pr` and saving the result as an object called `pr.var`.

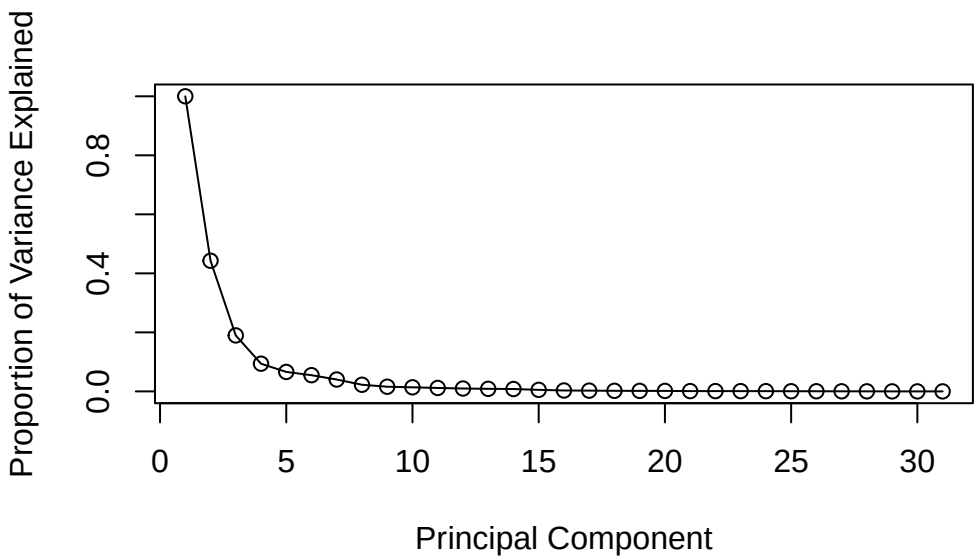
```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

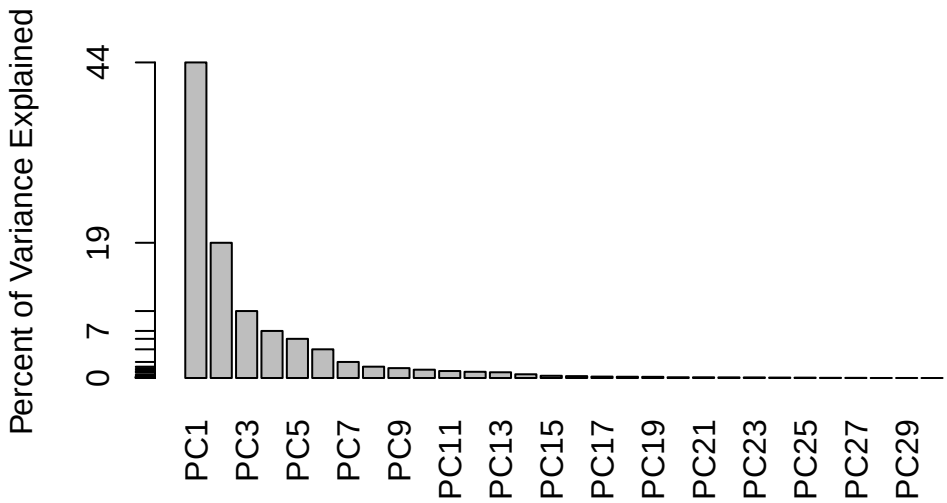
We will calculate the variance explained by each PC by dividing it by the total variance of all PCs. We will assign this to a variable called `pve` and make a plot of variance explained by each PC.

```r
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```r
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation[,1]
```

```
            radius_mean           texture_mean         perimeter_mean
            -0.21890244            -0.10372458            -0.22753729
              area_mean        smoothness_mean       compactness_mean
            -0.22099499            -0.14258969            -0.23928535
         concavity_mean    concave.points_mean          symmetry_mean
            -0.25840048            -0.26085376            -0.13816696
 fractal_dimension_mean              radius_se             texture_se
            -0.06436335            -0.20597878            -0.01742803
           perimeter_se                area_se          smoothness_se
            -0.21132592            -0.20286964            -0.01453145
         compactness_se           concavity_se      concave.points_se
            -0.17039345            -0.15358979            -0.18341740
```

```
           symmetry_se    fractal_dimension_se            radius_worst
           -0.04249842             -0.10256832             -0.22799663
         texture_worst         perimeter_worst              area_worst
           -0.10446933             -0.23663968             -0.22487053
      smoothness_worst       compactness_worst         concavity_worst
           -0.12795256             -0.21009588             -0.22876753
  concave.points_worst          symmetry_worst fractal_dimension_worst
           -0.25088597             -0.12290456             -0.13178394
```

The component of the loading vector for the feature `concave.points_mean` is -0.26085376. There are no features with larger contributions than this one.

Hierarchical Clustering

In order to prepare for hierarchical clustering, we need to create a distance matrix. First let's scale the `wisc.data` data and assign the result to `data.scaled`.

```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
```

Next we'll calculate the distances between all pairs of observations in the scaled dataset and store the result in `data.dist`.

```
data.dist <- dist(data.scaled)
```

We'll create a hierarchical clustering model using `hclust()` and complete linkage and assign the result to `wisc.hclust`.

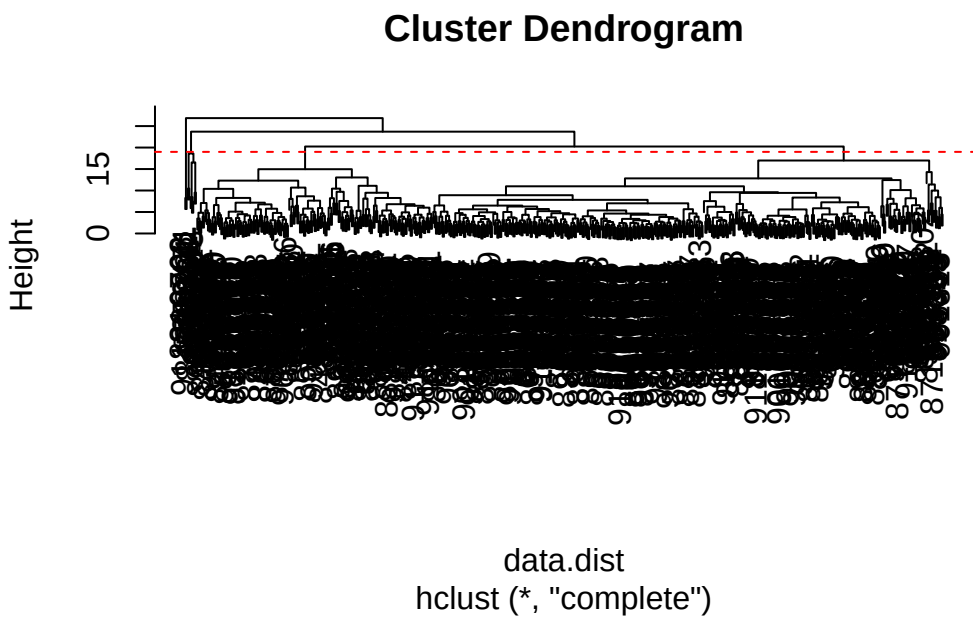
```
wisc.hclust <- hclust(data.dist, method="complete")
```

Results of hierarchical clustering

Using the model we just made, we'll determine a height where a certain number of clusters exists.

> Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```r
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```



At a height of 19, the clustering model has 4 clusters.

Selecting number of clusters

We'll use `cutree()` to cut the tree into 4 clusters. The output will be assigned to the variable `wisc.hclust.clusters`.

```r
wisc.hclust.clusters <- cutree(wisc.hclust, h=19)
```

We'll use the `table()` function to compare the cluster membership to the actual diagnoses.

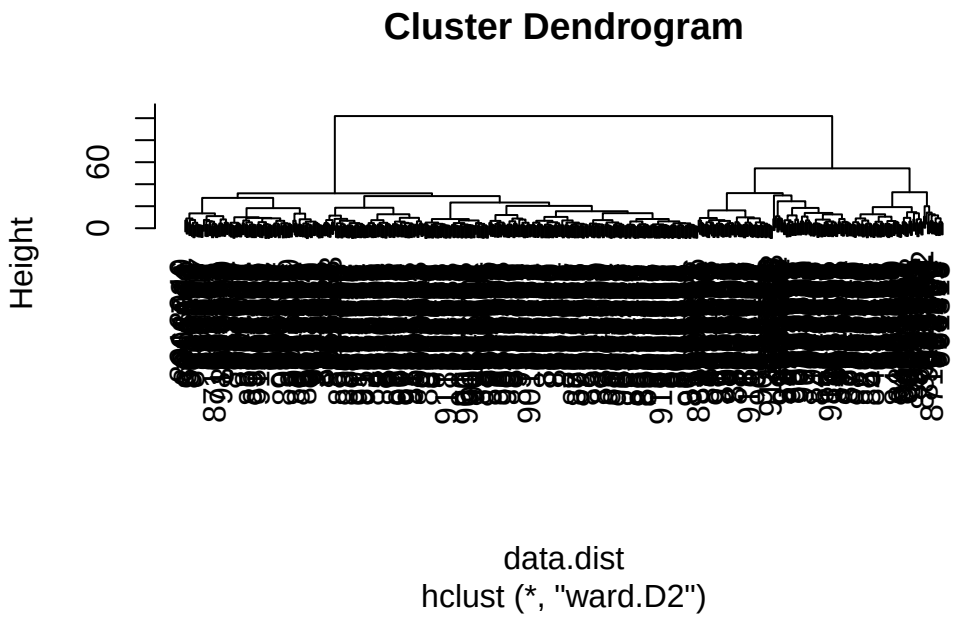
```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Using different methods

> Q12. Which method gives your favorite results for the same `data.dist` dataset? Explain your reasoning.

```
wisc.hclust <- hclust(data.dist, method="ward.D2")
plot(wisc.hclust)
```



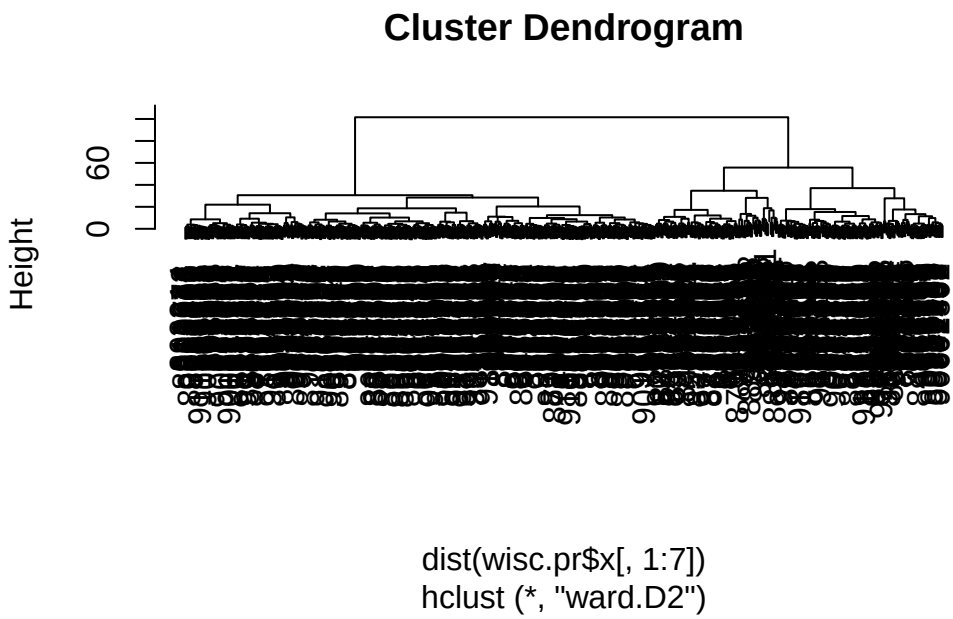
`ward.D2` gives my favorite results for the `data.dist` dataset, because the "tree branches" are more evenly split and the clusters are easier to visualize.

Combining methods

Clustering on PCA results

Using 7 PCs (describes at least 90% of the variability in the data), we will create a hierarchical clustering model with the linkage method `ward.D2`. We will assign the output to the variable `wisc.pr.hclust`.

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:7]), method="ward.D2")
plot(wisc.pr.hclust)
```



There are two main clusters on the dendrogram. Let's determine if these represent malignant and benign.

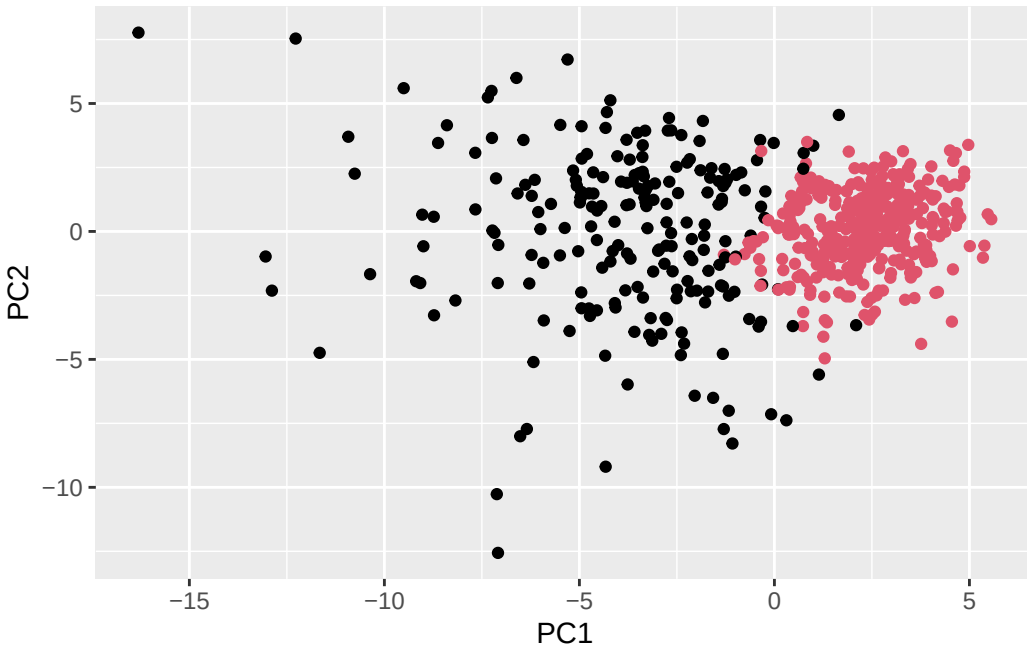
```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
216 353
```

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



We'll cut this model into 2 clusters and store the output in `wisc.pr.hclust.clusters`.

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
```

Using `table()`, compare the results from the new model with the actual diagnoses.

> Q13. How well does the newly created `hclust` model with two clusters separate out the two "M" and "B" diagnoses?

```
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

The newly created model is very accurate with separating the "M" and "B" diagnoses. There is a small amount of error, but for the most part it is correct.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

This model does a fairly good job at separating the diagnoses, but not as good as `wisc.pr.hclust.clusters`. The number of clusters also does not match for `wisc.hclust.clusters`.

Prediction

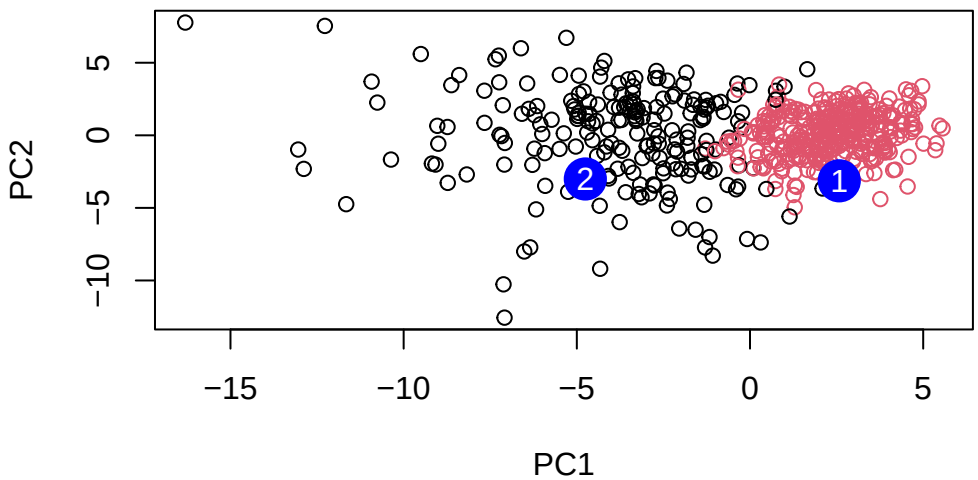
Using the `predict()` function we will take our PCA model and new cancer cell data and project the new data onto our PCA space.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
```

```
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize patient 2 for follow up because they are located within the malignant cluster.